UGANDA REVISION HUB

S6 Starter Study Notes

Batch 1 • Online and offline revision

S6 • Mathematics

Differentiation from first principles and basic rules

Learning objectives

- Interpret a derivative as a gradient or rate of change
- Differentiate simple powers of x
- Evaluate a derivative at a given point

Lesson notes

Differentiation measures how a function changes. Geometrically, the derivative at a point gives the gradient of the tangent to the curve there; in applications it can represent an instantaneous rate of change.

For a power function y = ax■, the power rule gives dy/dx = anx■■¹. A constant differentiates to zero, and the derivative of a sum is the sum of the derivatives.

Always simplify carefully and remember that differentiation reduces the power by one while multiplying by the original power.

Worked example

If y = 3x² + 4x - 7, then dy/dx = 6x + 4. At x = 2, the gradient is 6(2)+4 = 16.

Revision questions

1. Differentiate y = 5x³.
2. Differentiate y = 4x² - 3x + 8.
3. Find dy/dx for y = x² at x = 3.

Suggested answers

1. 15x²
2. 8x - 3
3. dy/dx = 2x, so the gradient is 6

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S6 • Biology

Genetics: genes, alleles and inheritance

Learning objectives

- Distinguish genes and alleles
- Explain genotype and phenotype
- Use a simple genetic cross to predict outcomes

Lesson notes

A gene is a section of DNA associated with a characteristic or biological function. Alleles are different forms of the same gene. In diploid organisms, an individual generally has two alleles for each autosomal gene, one inherited from each parent.

The genotype is the allele combination an organism carries, such as TT, Tt or tt. The phenotype is the observable characteristic produced by the genotype and the environment.

In a simple dominant-recessive model, a dominant allele can mask the effect of a recessive allele in a heterozygote. Real inheritance can be more complex, and not all traits follow this pattern.

Worked example

For T = tall (dominant) and t = short (recessive), Tt × Tt can produce TT, Tt, Tt and tt. The expected genotype ratio is 1:2:1 and, under this simple model, the phenotype ratio is 3 tall : 1 short.

Revision questions

1. What is an allele?
2. State the difference between genotype and phenotype.
3. For Tt × Tt, what fraction of offspring are expected to be tt?

Suggested answers

1. An alternative form of a gene
2. Genotype is the allele combination; phenotype is the observable characteristic
3. 1/4 or 25%

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